

ADITYA INDURKHYA

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SUMMARY

Data Engineer with **2+ years** of production experience building ETL/ELT pipelines, data warehouses, and reporting solutions, and an **M.S. in Data Analytics** from George Mason University. Scaled pipelines to **12M+ records monthly**, cut p95 query latency **from 35s to 18s**, and improved data reliability through automated validation and governance. Skilled in **Python, SQL, cloud data warehouses, and BI tools**.

TECHNICAL SKILLS

Programming	Python (pandas, NumPy), SQL, R, C/C++
Data Engineering	ETL/ELT Pipelines, Data Warehousing, Data Modeling, Dimensional Modeling, SQL Optimization, Data Quality, Data Governance
Big Data & Processing	Apache Spark (PySpark), Databricks, Snowflake, Batch Processing, Partitioning, Indexing
Cloud & Dev Tools	AWS (S3, Redshift, RDS, Glue), Microsoft Azure, Docker, FastAPI, REST APIs, Git/GitHub, CI/CD
Databases	PostgreSQL (PostGIS), Amazon Redshift, MySQL, MongoDB
BI & Visualization	Power BI (DAX), Tableau, Apache Superset, Matplotlib, Plotly, ggplot2
Analytics & ML	Statistical Analysis, Feature Engineering, Regression, Classification, Clustering, Time Series Forecasting (ARIMA), A/B Testing, Hypothesis Testing, Deep Learning (CNN)

PROFESSIONAL EXPERIENCE

Data Engineer — ICSS, Jabalpur, India Jul 2022 – Aug 2023

32-year-old IT hardware supplier specializing in large-scale government procurement. Owned the company's first unified data warehouse implementation.

- Engineered and scaled **PySpark** ETL pipelines processing **12M+ records monthly** across **8 heterogeneous sources** for the company's first unified data warehouse, cutting daily execution **from 2.0 to 1.4 hours** at **95%+ uptime**.
- Optimized **PostgreSQL and Redshift** performance on **50GB+** datasets by redesigning indexes and rewriting hot-path SQL, reducing p95 query latency **from 35s to 18s** and moving reporting from same-afternoon to **same-morning** for the finance and operations team.
- Automated data validation with a **Python** framework and anomaly alerting, reducing pipeline data issues **from 12% to 5%** of daily runs and saving **15 staff-hours weekly**; documented governance standards across **5 design documents** adopted by the engineering team.

Data Engineering Intern — D'Festo Private Limited, Jabalpur, India May 2021 – May 2022

Sub-10-person early-stage company running operations entirely on manual spreadsheets, with no existing analytical infrastructure.

- Established the company's first automated **ETL and reporting system** using **Python, pandas, and PostgreSQL**, processing **2M+ records daily** across 3 source systems with built-in validation and error handling.
- Improved data reliability by auditing existing reporting paths and resolving **20 data quality defects** (duplicates, nulls, key violations); reduced recurring query times **40%** through targeted **PostgreSQL indexing**.
- Delivered **4 Power BI dashboards** with **DAX**-based KPIs replacing manual spreadsheet tracking, and wrote runbooks enabling non-engineers to triage **10 monthly ingestion alerts** without escalation.

PROJECTS

ClimateAI — Geospatial Climate Data Platform | M.S. Capstone Jan 2026 – May 2026

- Led data engineering for a **6-member team**, architecting geospatial **ETL pipelines** and a **PostgreSQL/PostGIS** warehouse ingesting **250K+ climate records** at **95%+ completeness**.
- Optimized **spatial indexing and table partitioning** to **sub-300ms** query response; implemented an automated validation layer (missing values, invalid coordinates, outliers) with pipeline-health monitoring at **96%+ uptime**.
- Deployed **Dockerized FastAPI** services exposing SQL-backed retrieval with structured request validation, sustaining a **99% query success rate** across stakeholder demos.

Employee Attrition Prediction — Workforce Analytics Platform Jan 2025 – May 2025

- Engineered a **SQL and Python** analytics pipeline over **8,000+ multi-year employee records**, engineering **25+ features** (tenure, salary history, promotion gaps, department transfers).
- Applied **statistical analysis** (t-tests, chi-square) and predictive modeling (**$R^2 = 0.65$, 80%+ accuracy**) to identify the **top 5 attrition drivers** across department and tenure cohorts.
- Delivered **Power BI** dashboards with attrition segmentation and employee risk scoring that informed HR workforce-planning decisions.

Pavement Deterioration Analysis — Infrastructure Maintenance Modeling Aug 2024 – Dec 2024

- Developed **Python** data-cleaning pipelines (**pandas, scikit-learn, Statsmodels**) preparing **18,000+ inconsistent inspection entries** across 3 years, deriving **30+ features** spanning weather, traffic, and maintenance history.
- Modeled deterioration drivers via **multiple linear regression and one-way ANOVA**, confirming pavement type as statistically significant (**$p = 0.0013$**); validated model limits via residual and Q-Q analysis.
- Co-authored a research paper and presented findings at a **regional engineering conference**, informing infrastructure maintenance planning.

EDUCATION

M.S. Data Analytics — George Mason University, Fairfax, VA | GPA: 3.73/4.0 Aug 2024 – May 2026

B.Tech. Computer Science & Engineering — Gyan Ganga Institute of Technology and Sciences, India Aug 2020 – Jun 2024

CERTIFICATIONS

AWS Academy: Data Engineering (coursework) | LinkedIn Learning: SQL for Data Analysis, Learning SQL Programming | DataCamp: Data Manipulation with dplyr, Modeling with tidymodels